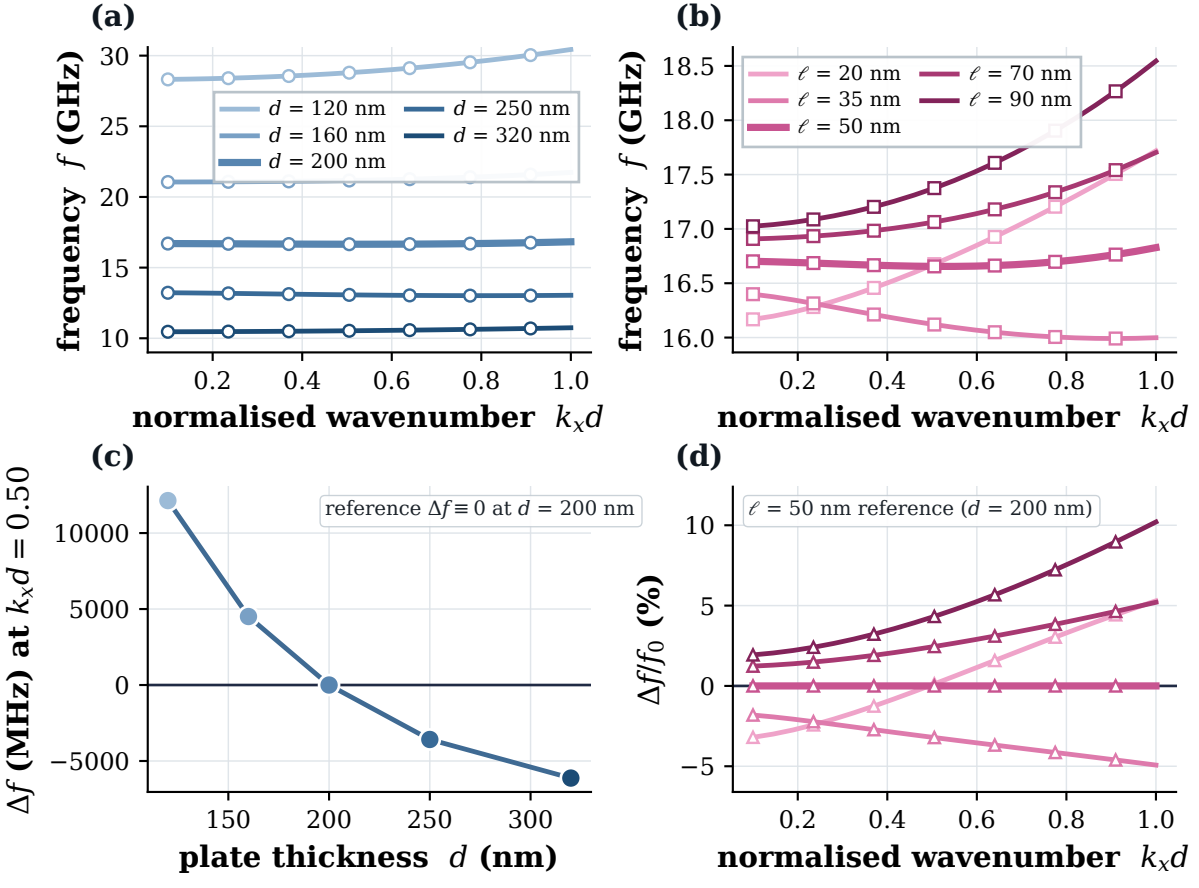
(a)
frequency f (GHz)
30
25
20
15
10
d = 120 nm
d = 160 nm
d = 200 nm
d = 250 nm
d = 320 nm
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(b)
frequency f (GHz)
18.5
18.0
17.5
17.0
16.5
16.0
ℓ = 20 nm
ℓ = 35 nm
ℓ = 50 nm
ℓ = 70 nm
ℓ = 90 nm
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(c)
Δf (MHz) at $k_x d = 0.50$
reference $\Delta f \equiv 0$ at d = 200 nm
10000
5000
0
−5000
150
200
250
300
plate thickness d (nm)
(d)
$\Delta f / f_0$ (%)
ℓ = 50 nm reference (d = 200 nm)
10
5
0
−5
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$